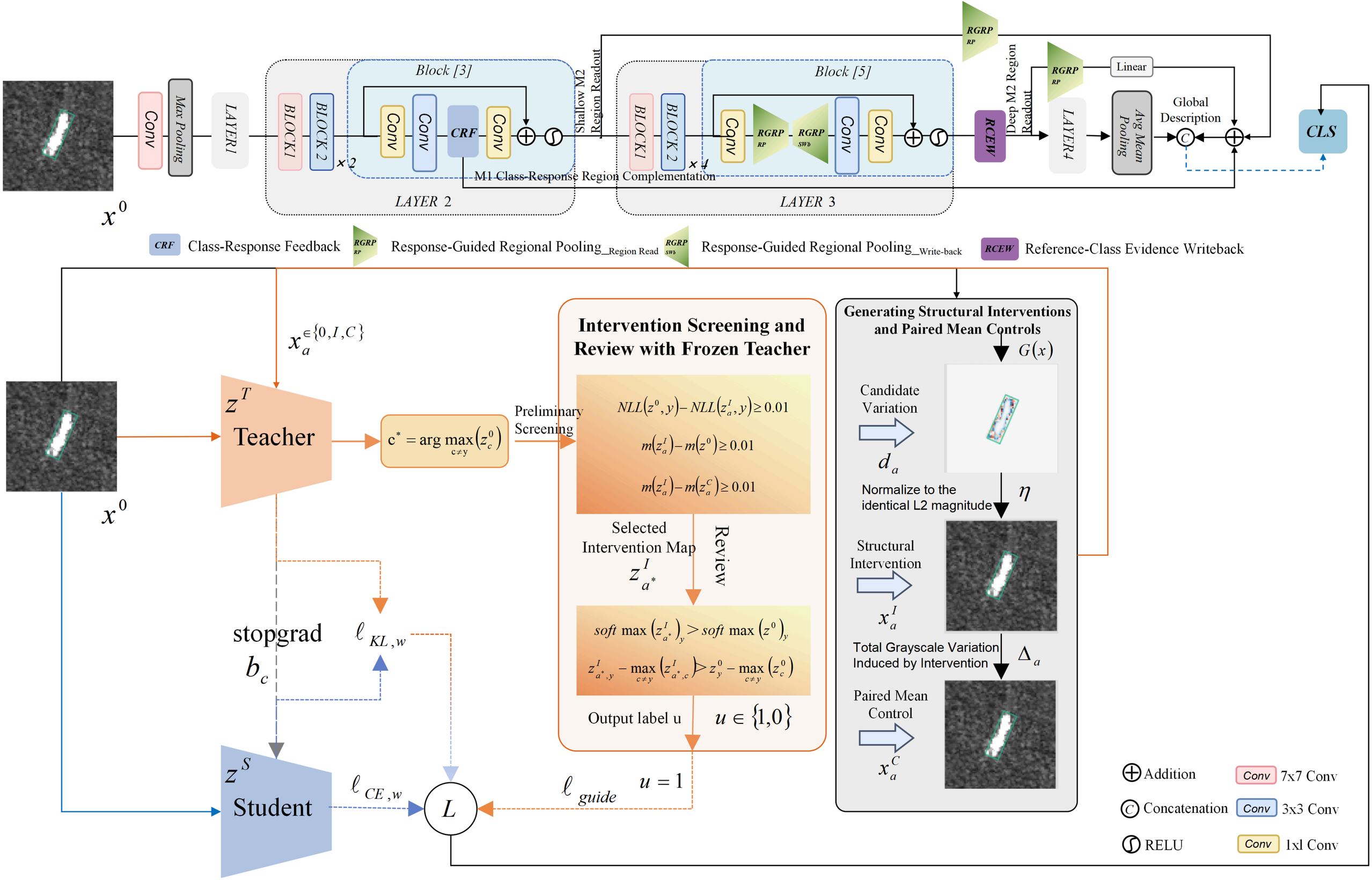

Conv
Max Pooling
LAYER1
BLOCK1
BLOCK 2
× 2
Block [3]
Conv
Conv
CRF
Conv
Shallow M2 Region Readout
M1 Class-Response Region Complementation
LAYER 2
BLOCK1
BLOCK 2
× 4
Block [5]
Conv
RGRP RP
RGRP SWb
Conv
Conv
LAYER 3
RCEW
Deep M2 Region Readout
RGRP RP
LAYER4
Linear
Avg Mean Pooling
Global Description
CLS
x^0
CRF Class-Response Feedback
RGRP RP Response-Guided Regional Pooling_Region Read
RGRP SWb Response-Guided Regional Pooling_Write-back
RCEW Reference-Class Evidence Writeback
$x_a^{\in\{0,I,C\}}$
z^T
Teacher
x^0
$c^* = \arg\max_{c\neq y}(z_c^0)$
Preliminary Screening
Intervention Screening and Review with Frozen Teacher
$NLL(z^0,y) - NLL(z_a^I,y) \geq 0.01$
$m(z_a^I) - m(z^0) \geq 0.01$
$m(z_a^I) - m(z_a^C) \geq 0.01$
Selected Intervention Map
$z_{a^*}^I$
Review
$soft\ \max(z_{a^*}^I)_y > soft\ \max(z^0)_y$
$z_{a^*,y}^I - \max_{c\neq y}(z_{a^*,c}^I) > z_y^0 - \max_{c\neq y}(z_c^0)$
Output label u
$u \in \{1,0\}$
Generating Structural Interventions and Paired Mean Controls
$G(x)$
Candidate Variation
d_a
Normalize to the identical L2 magnitude
η
Structural Intervention
x_a^I
Total Grayscale Variation Induced by Intervention
Δ_a
Paired Mean Control
x_a^C
stopgrad
b_c
$\ell_{KL,w}$
z^S
Student
$\ell_{CE,w}$
L
ℓ_{guide}
$u = 1$
Addition
Concatenation
RELU
Conv 7x7 Conv
Conv 3x3 Conv
Conv 1xl Conv